

Cross-dataset 4 Results and Analysis

We train and test our dataset for:

GRU (Gated Recurrent Unit), Transformer and GAN (Generative Adversarial Network)

Train Dataset: trainset_synth.csv

Test Dataset: feb2026rb2r2_updated.txt

Summary:

For this dataset, all models struggle, with negative R^2 for both voltage and power, indicating poor generalization. For voltage, Transformer performs slightly better than GRU, while GAN performs significantly worse. For power, GRU shows relatively better performance compared to Transformer and GAN, though still unreliable. **Overall, temporal and attention models degrade under this dataset, while GAN remains the weakest, suggesting strong distribution shift or data complexity affecting all models.**

Voltage GRU Results:

MAE: 0.31701430503415107

RMSE: 0.4414703278801313

R2: -0.7206901102676562

Power GRU Results:

MAE: 81.92084850902495

RMSE: 148.84174524292214

R2: -0.11574685020564668

Voltage Transformer Results:

MAE: 0.29818284929746997

RMSE: 0.43332460180134325

R2: -0.6577685812651608

Power Transformer Results:

MAE: 117.19309579122933

RMSE: 172.58258915745387

R2: -0.5000386833528767

Voltage GAN Results:

MAE: 0.9110893124655886

RMSE: 0.9837452021903079

R2: -7.5440241218726545

Power GAN Results:

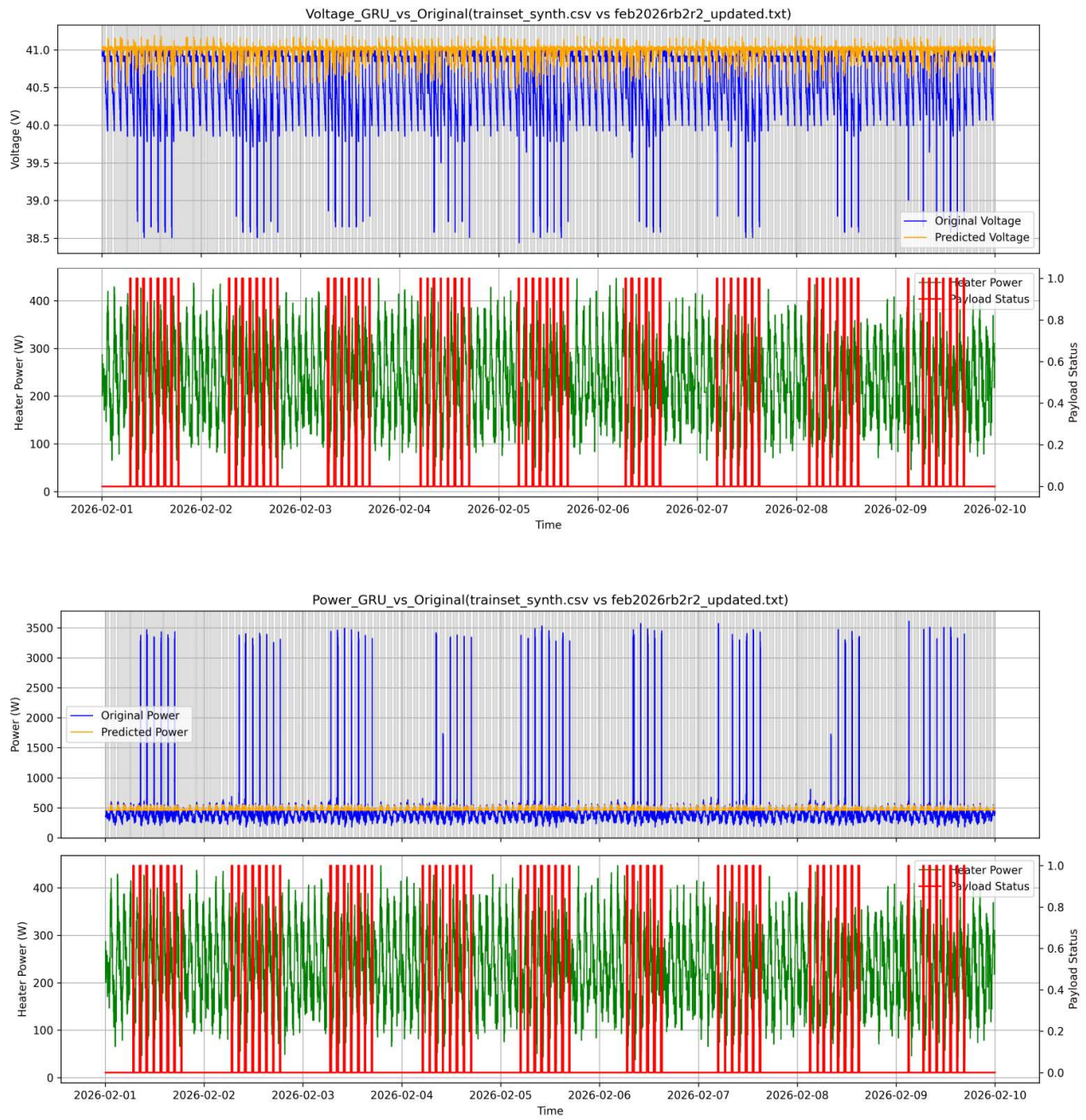
MAE: 137.67170632683218

RMSE: 187.80385930526455

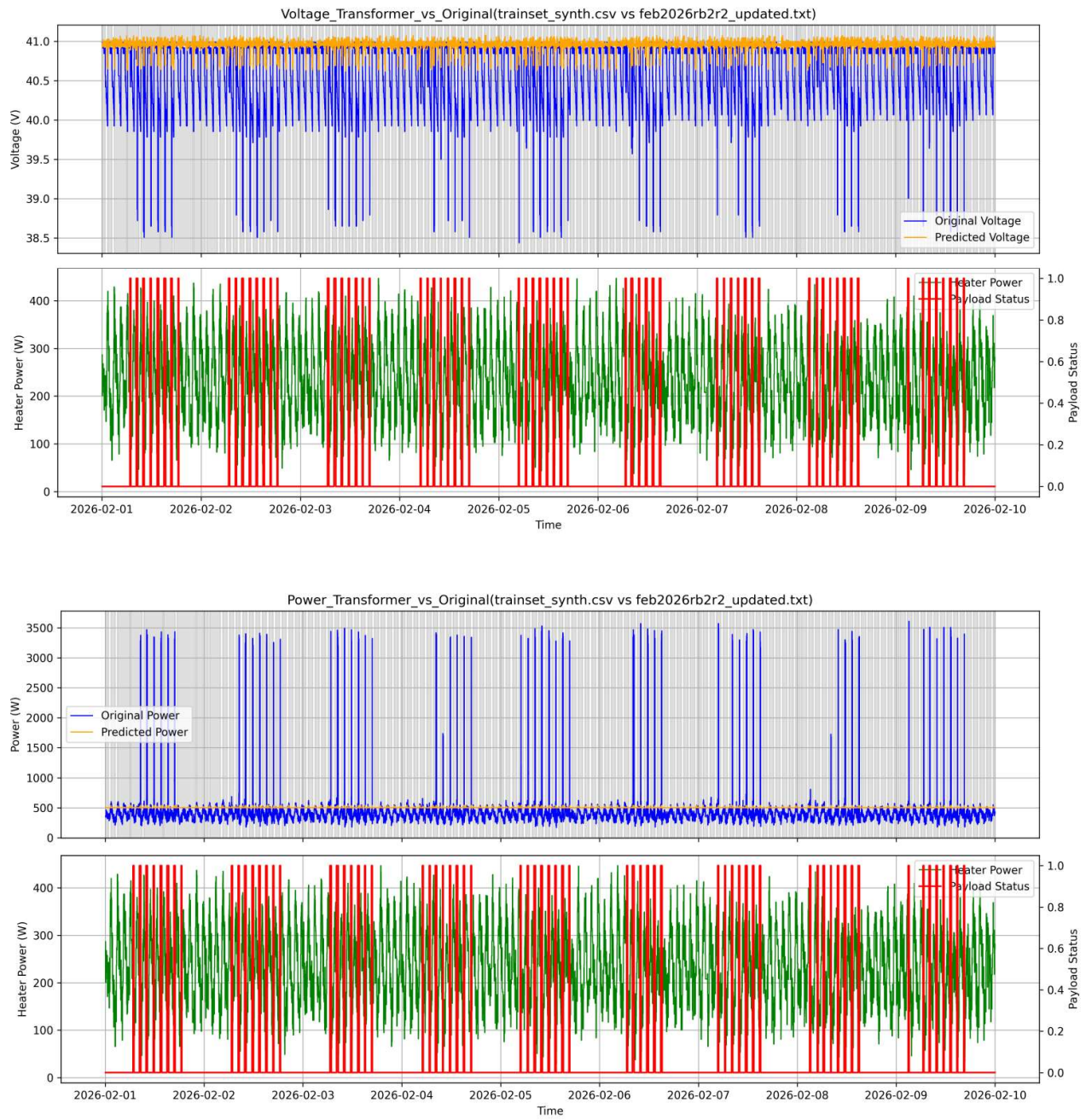
R2: -0.7763049407956915

Plots:

1. GRU



2. Transformer



3. GAN

